

MÔN HỌC: CƠ SỞ DỮ LIỆU NÂNG CAO

BÀI THỰC HÀNH 4: CƠ SỞ DỮ LIỆU PHÂN BỐ

PHÂN MẢNH NGANG, PHÂN MẢNH DỌC CÁC BẢNG

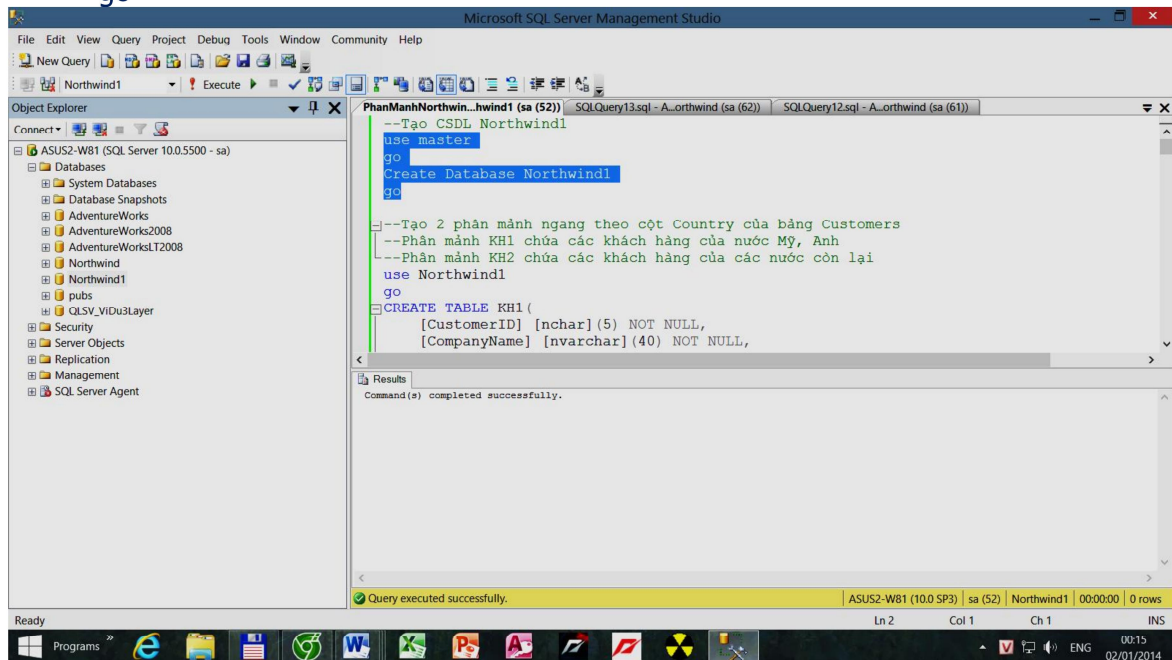
(HƯỚNG DẪN GIẢI PHẦN 1 LÀM TẠI LỚP)

Chú ý: bài thực hành sử dụng CSDL mẫu Northwind của Microsoft, sinh viên nên làm tại phòng thực hành tất cả các câu . Bài tập về nhà buổi 4 cho riêng file khác.

Phân mảnh ngang bảng:

1. Tạo CSDL Northwind1:

```
use master
go
Create Database Northwind1
go
```



2. Tạo 2 phân mảnh ngang theo cột Country của bảng Customers:

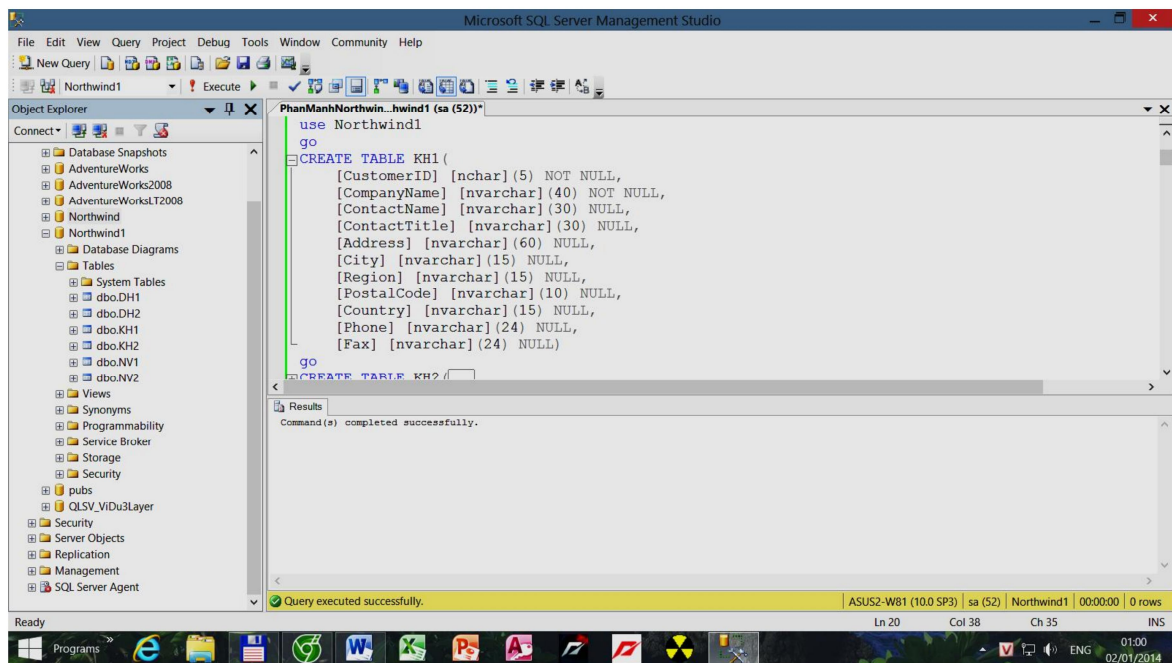
```
-Phân mảnh KH1 chứa các khách hàng của nước Mỹ, Anh
-Phân mảnh KH2 chứa các khách hàng của các nước còn lại
use Northwind1
go
CREATE TABLE KH1(
    [CustomerID] [nchar](5) NOT NULL,
```

```

[CompanyName] [nvarchar](40) NOT NULL,
[ContactName] [nvarchar](30) NULL,
[ContactTitle] [nvarchar](30) NULL,
[Address] [nvarchar](60) NULL,
[City] [nvarchar](15) NULL,
[Region] [nvarchar](15) NULL,
[PostalCode] [nvarchar](10) NULL,
[Country] [nvarchar](15) NULL,
[Phone] [nvarchar](24) NULL,
[Fax] [nvarchar](24) NULL

```

go



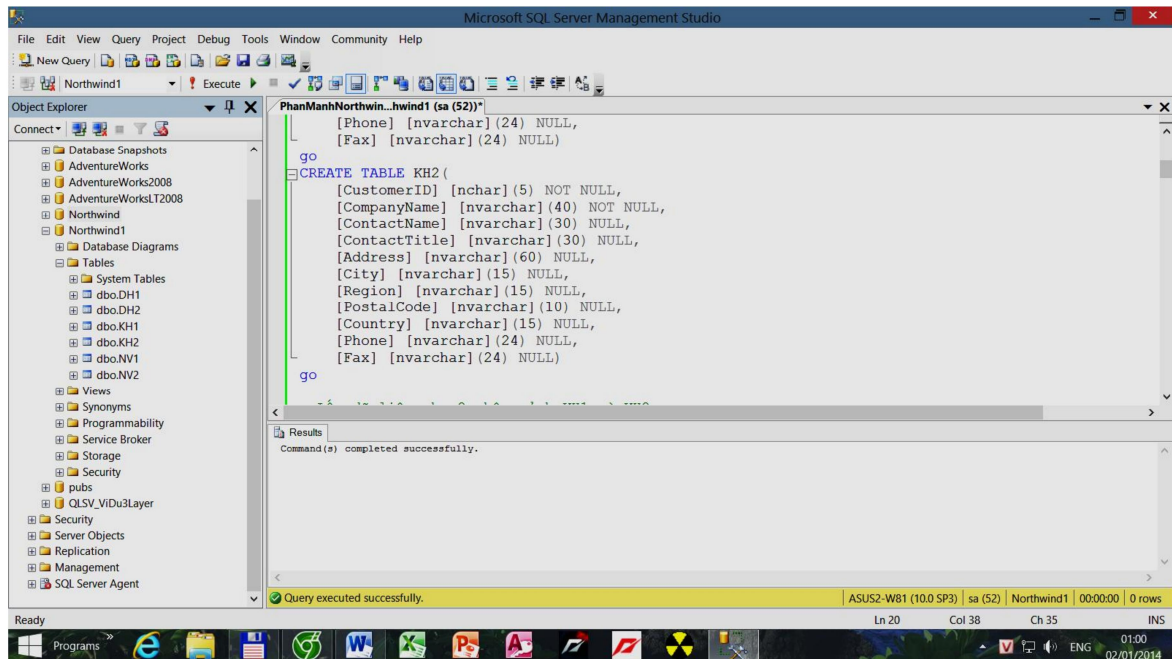
CREATE TABLE KH2(

```

[CustomerID] [nchar](5) NOT NULL,
[CompanyName] [nvarchar](40) NOT NULL,
[ContactName] [nvarchar](30) NULL,
[ContactTitle] [nvarchar](30) NULL,
[Address] [nvarchar](60) NULL,
[City] [nvarchar](15) NULL,
[Region] [nvarchar](15) NULL,
[PostalCode] [nvarchar](10) NULL,
[Country] [nvarchar](15) NULL,
[Phone] [nvarchar](24) NULL,
[Fax] [nvarchar](24) NULL

```

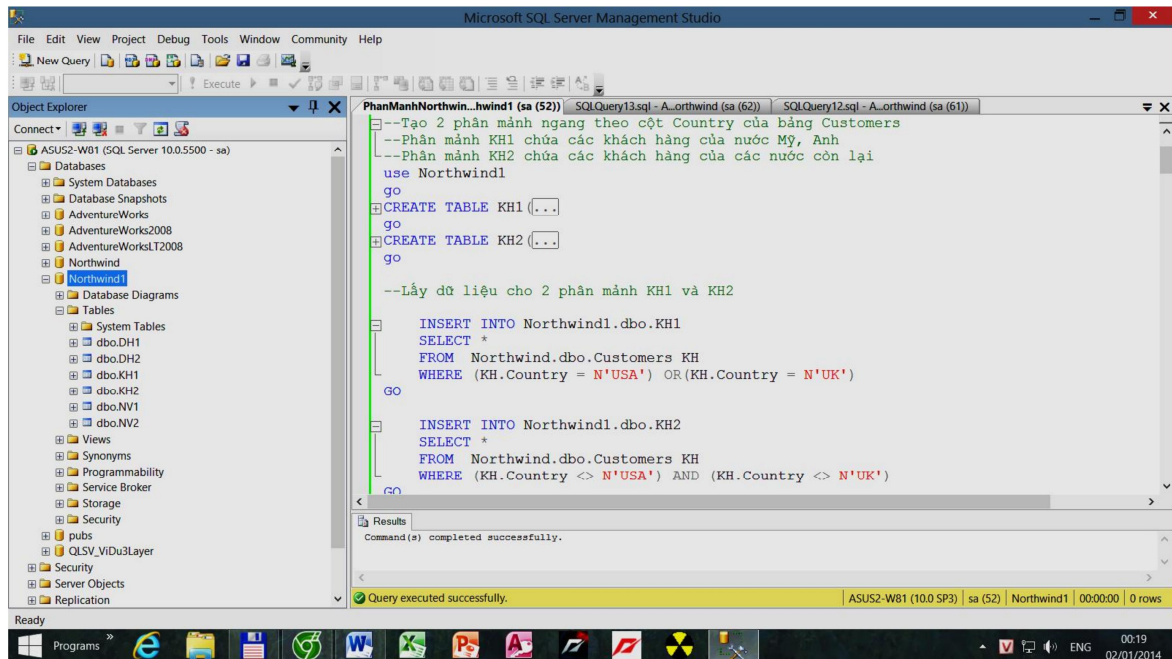
go



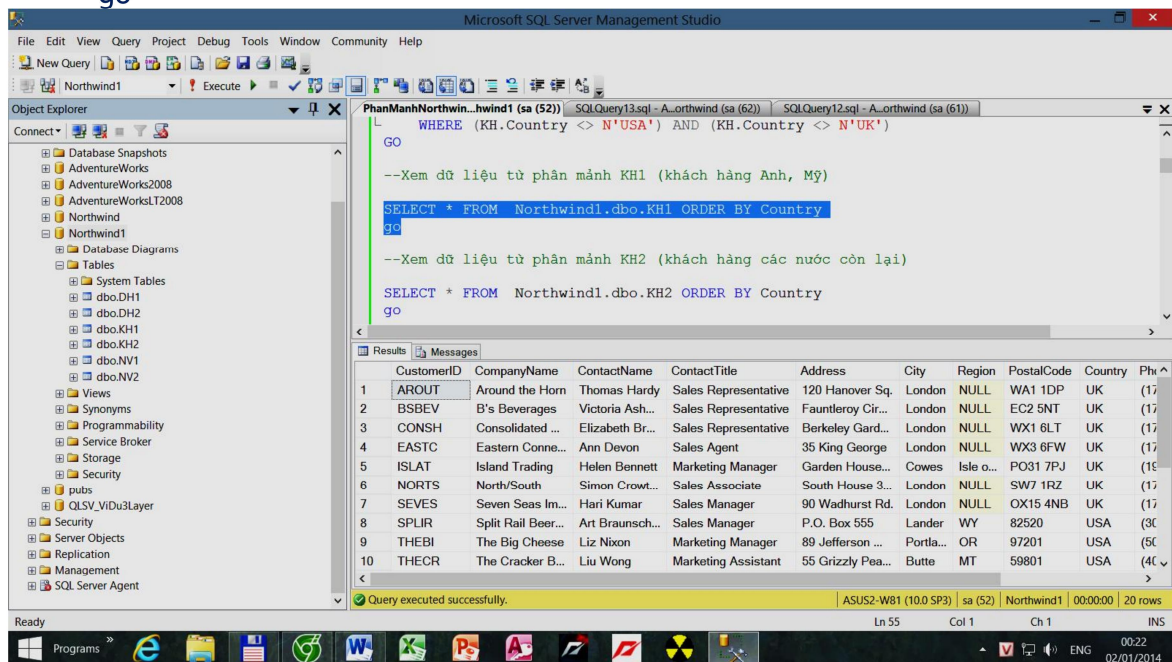
-Lấy dữ liệu cho 2 phân mảnh KH1 và KH2

```
INSERT INTO Northwind1.dbo.KH1
SELECT *
FROM Northwind.dbo.Customers KH
WHERE (KH.Country = N'USA') OR (KH.Country = N'UK')
GO
```

```
INSERT INTO Northwind1.dbo.KH2
SELECT *
FROM Northwind.dbo.Customers KH
WHERE (KH.Country <> N'USA') AND (KH.Country <> N'UK')
GO
```



3. Xem dữ liệu từ phân mảnh KH1 (khách hàng Anh, Mỹ)
- ```
SELECT * FROM Northwind1.dbo.KH1 ORDER BY Country
go
```



4. Xem dữ liệu từ phân mảnh KH2 (khách hàng các nước còn lại)
- ```
SELECT * FROM Northwind1.dbo.KH2 ORDER BY Country
go
```

Microsoft SQL Server Management Studio

Object Explorer: Northwind1

Query: PhanManhNorthwin...hwind1 (sa (52))

```
--Xem dữ liệu từ phân mảnh KH1 (khách hàng Anh, Mỹ)
SELECT * FROM Northwind1.dbo.KH1 ORDER BY Country
go
--Xem dữ liệu từ phân mảnh KH2 (khách hàng các nước còn lại)
SELECT * FROM Northwind1.dbo.KH2 ORDER BY Country
go
--Lấy danh sách tất cả khách hàng:
--Truy vấn trong suốt mức 1 (hay mức toàn cục, mức không thấy các phân mảnh,
-- hay mức trong suốt phân mảnh, fragmentation transparency)
```

anyName	ContactName	ContactTitle	Address	City	Region	PostalCode	Country	Phone
1	s Comidas para Il...	Patricio Simpson	Sales Agent	Cerrito 333	Buenos Aires	NULL	1010	Argentina (1) 135-555
2	o Atlántico Ltda.	Yvonne Moncada	Sales Agent	Ing. Gustavo Moncada 8...	Buenos Aires	NULL	1010	Argentina (1) 123-555
3	o grande	Sergio Gutiérrez	Sales Representative	Av. del Libertador 900	Buenos Aires	NULL	1010	Argentina (1) 123-555
4	o und mehr	Georg Pippes	Sales Manager	Geisweg 14	Salzburg	NULL	5020	Austria 6562-9722
5	Handel	Roland Mendel	Sales Manager	Kirchgasse 6	Graz	NULL	8010	Austria 7675-3425
6	n Dewey	Catherine Dewey	Sales Agent	Rue Joseph-Bens 532	Bruxelles	NULL	B-1180	Belgium (02) 201 24
7	mes d'éclopes	Pascale Cartrain	Accounting Manager	Boulevard Tirou, 255	Charleroi	NULL	B-6000	Belgium (071) 23 67
8	o Adocicados	Janete Limeira	Assistant Sales Agent	Av. Copacabana, 267	Rio de Janeiro	RJ	02389-890	Brazil (21) 555-34
9	ão Hipermercados	Anabela Domingues	Sales Representative	Av. Inês de Castro, 414	Sao Paulo	SP	05634-030	Brazil (11) 555-21
10	ntro Imortadva	Paula Parente	Sales Manager	Rua do Mercado, 12	Resende	SP	08737-363	Brazil (14) 555-81

Query executed successfully.

ASUS2-W81 (10.0 SP3) | sa (52) | Northwind1 | 00:00:00 | 71 rows

- Lấy danh sách tất cả khách hàng: Truy vấn trong suốt mức 1 (hay mức toàn cục, mức Global, mức không thấy các phân mảnh, hay mức trong suốt phân mảnh, fragmentation transparency)

```
USE Northwind
```

```
GO
```

```
CREATE PROC DStatCaKHMUC1
```

```
AS
```

```
SELECT * FROM Northwind.dbo.Customers
```

```
go
```

```
EXEC DStatCaKHMUC1
```

```
go
```

```
DROP PROC DStatCaKHMUC1
```

```
GO
```

The screenshot shows the Microsoft SQL Server Management Studio interface. The query window contains the following T-SQL code:

```

-- hay mức trong suốt phân mảnh, fragmentation transparency)
USE Northwind
GO
CREATE PROC DSTatCaKHMUC1
AS
SELECT * FROM Northwind.dbo.Customers
go
EXEC DSTatCaKHMUC1
go
DROP PROC DSTatCaKHMUC1
GO
--Truy vấn trong suốt mức 2 (mức thấy các phân mảnh, không thấy các mảnh ở phần cục bộ,
-- location transparency)

```

The results grid displays the following data:

CustomerID	CompanyName	ContactName	ContactTitle	Address	City	Region	P
1	ALFKI	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	NULL	1
2	ANATR	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	NULL	0
3	ANTON	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	NULL	0
4	AROUT	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	NULL	V
5	BERGS	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	NULL	S
6	BLAUS	Blauer See Delikatessen	Hanna Moos	Forsterstr. 57	Mannheim	NULL	6
7	BLONP	Blondesddsl père et fils	Frédérique Citeaux	24, place Kléber	Strasbourg	NULL	6
8	BOLID	Bólido Comidas preparadas	Martín Sommer	C/ Araquil, 67	Madrid	NULL	2
9	BONAP	Bon app'	Laurence Leblan	12, rue des Bouchers	Marseille	NULL	1
10	BOTTM	Bottm-Dollar Markets	Elizabeth Lincoln	23 Tsawassen Blvd	Tsawassen	BC	T

The status bar at the bottom indicates: Query executed successfully. | ASUS2-WB1 (10.0 SP3) | sa (52) | Northwind | 00:00:00 | 91 rows

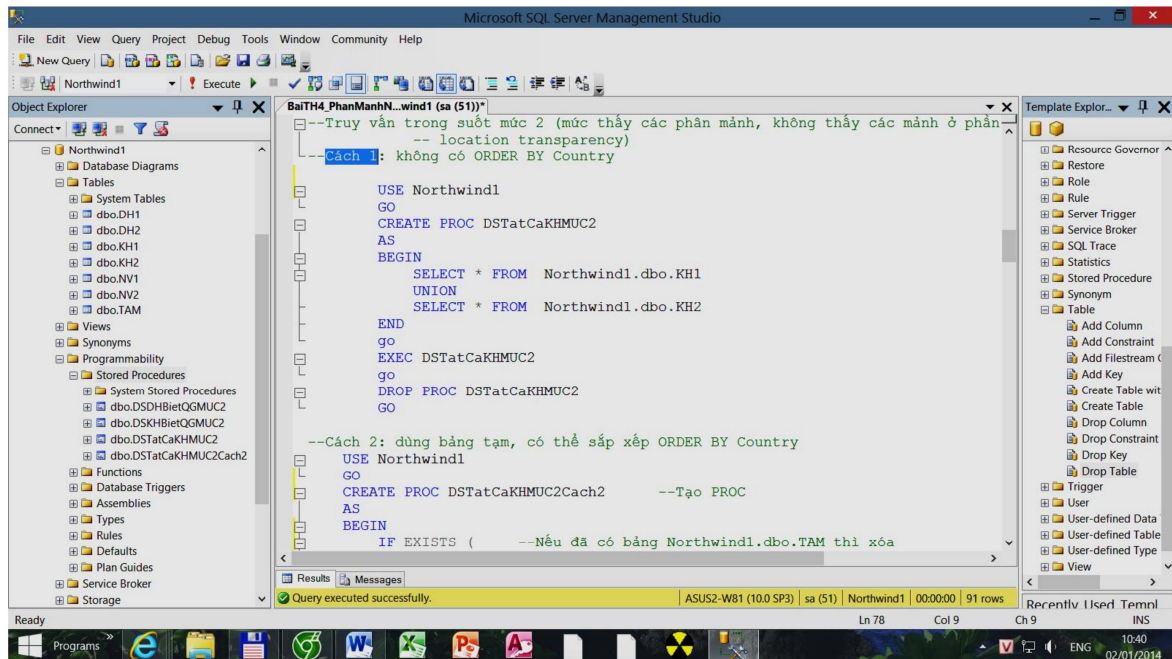
6. Lấy danh sách tất cả khách hàng: Truy vấn trong suốt mức 2 (mức thấy các phân mảnh, không thấy các mảnh ở phần cục bộ, location transparency)

--Cách 1: không có ORDER BY Country

```

USE Northwind1
GO
CREATE PROC DSTatCaKHMUC2
AS
BEGIN
    SELECT * FROM Northwind1.dbo.KH1
    UNION
    SELECT * FROM Northwind1.dbo.KH2
END
go
EXEC DSTatCaKHMUC2
go
DROP PROC DSTatCaKHMUC2
GO

```



--Cách 2: dùng bảng tạm, có thể sắp xếp ORDER BY Country

```
USE Northwind1
```

```
GO
```

--Tạo PROC

```
CREATE PROC DSTatCaKHMUC2Cach2
```

```
AS
```

```
BEGIN
```

--Nếu đã có bảngNorthwind1.dbo.TAM thì xóa:

```
IF EXISTS (
```

```
    SELECT *
```

```
    FROM sys.tables
```

```
    JOIN sys.schemas
```

```
    ON sys.tables.schema_id = sys.schemas.schema_id
```

```
    WHERE sys.schemas.name = 'dbo'
```

```
    AND sys.tables.name = 'TAM')
```

```
DROP TABLE Northwind1.dbo.TAM
```

--Vừa tạo vừa lấy dữ liệu cho bảng Northwind1.dbo.TAM từ phân mảnh KH1:

```
SELECT * INTO Northwind1.dbo.TAM
```

```
FROM Northwind1.dbo.KH1
```

--Lấy thêm dữ liệu từ phân mảnh KH2:

```
INSERT INTO Northwind1.dbo.TAM
```

```
    SELECT * FROM Northwind1.dbo.KH2
```

--Xuất kết quả từ bảng Northwind1.dbo.TAM

có sắp tăng dần theo quốc gia:

```
SELECT * FROM Northwind1.dbo.TAM ORDER BY Country
```

```
END
```

```
go
```

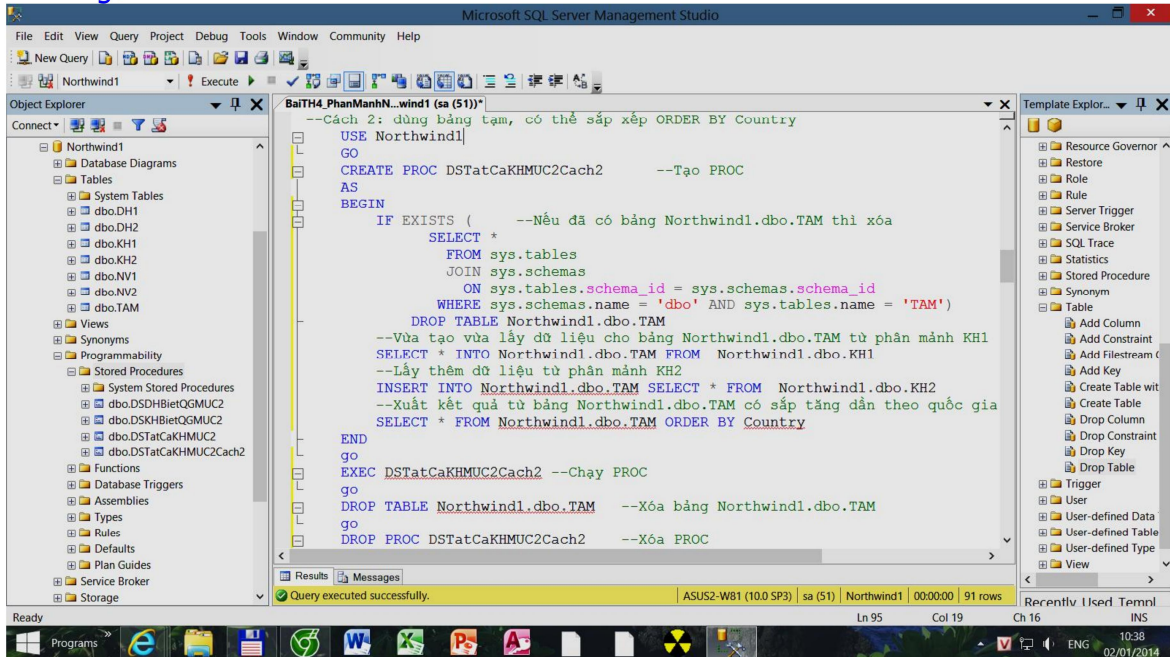
```
EXEC DSTatCaKHMUC2Cach2
```

--Chạy PROC

```

go
DROP TABLE Northwind1.dbo.TAM --Xóa bảng Northwind1.dbo.TAM
go
DROP PROC DSTatCaKHMUC2Cach2 --Xóa PROC
go

```



7. Danh sách khách hàng biết tên quốc gia: Mức 1:

```

use Northwind
go
CREATE PROC DSKHBietQGMUC1(@QG nvarchar(15)) --Danh sách KH biết
quốc gia mức 1
AS
BEGIN
    SELECT * FROM Northwind.dbo.Customers WHERE Country=@QG
END
go

EXEC DSKHBietQGMUC1 N'Canada'
go
EXEC DSKHBietQGMUC1 N'USA'
go
DROP PROC DSKHBietQGMUC1
GO

```

Microsoft SQL Server Management Studio

Object Explorer: Northwind database structure.

Query Editor (PhanManhNorthwin...rthwind (sa (52))):

```
--Danh sách khách hàng biết tên quốc gia:
--Mức 1:
use Northwind
go
CREATE PROC DSKHBietQGMUC1(@QG nvarchar(15)) --Danh sách KH biết quốc gia mức 1
AS
BEGIN
    SELECT * FROM Northwind.dbo.Customers WHERE Country=@QG
END
go

EXEC DSKHBietQGMUC1 N'Canada'
go
EXEC DSKHBietQGMUC1 N'USA'
go
```

Results:

CustomerID	CompanyName	ContactName	ContactTitle	Address	City	Region	PostalCode	Country	Phone
1	BOTTOM	Elizabeth Lincoln	Accounting Manager	23 Tsawassen Blvd.	Tsawassen	BC	T2F 8M4	Canada	(604) 291-9095
2	LAUGB	Yoshi Tannamuri	Marketing Assistant	1900 Oak St.	Vancouver	BC	V3F 2K1	Canada	(604) 261-9830
3	MEREP	Jean Fresnière	Marketing Assistant	43 rue St. Laurent	Montréal	Québec	H1J 1C3	Canada	(514) 358-5600

Query executed successfully. | ASUS2-W81 (10.0 SP3) | sa (52) | Northwind | 00:00:00 | 3 rows

Microsoft SQL Server Management Studio

Object Explorer: Northwind database structure.

Query Editor (PhanManhNorthwin...rthwind (sa (52))):

```
--Danh sách khách hàng biết tên quốc gia:
--Mức 1:
use Northwind
go
CREATE PROC DSKHBietQGMUC1(@QG nvarchar(15)) --Danh sách KH biết quốc gia mức 1
AS
BEGIN
    SELECT * FROM Northwind.dbo.Customers WHERE Country=@QG
END
go

EXEC DSKHBietQGMUC1 N'Canada'
go
EXEC DSKHBietQGMUC1 N'USA'
go
```

Results:

CustomerID	CompanyName	ContactName	ContactTitle	Address	City	Region
1	GREAL	Howard Snyder	Marketing Manager	2732 Baker Blvd.	Eugene	OR
2	HUNGC	Yoshi Latimer	Sales Representative	City Center Plaza 516 Main St.	Elgin	OR
3	LAZYK	John Steel	Marketing Manager	12 Orchestra Terrace	Walla Walla	WA
4	LETSS	Jaime Yorres	Owner	87 Polk St. Suite 5	San Francisco	CA
5	LONEP	Fran Wilson	Sales Manager	89 Chiaroscuro Rd.	Portland	OR
6	OLDWO	Rene Phillips	Sales Representative	2743 Bering St.	Anchorage	AK
7	RATTCC	Paula Wilson	Assistant Sales Representative	2817 Milton Dr.	Albuquerque	NM

Query executed successfully. | ASUS2-W81 (10.0 SP3) | sa (52) | Northwind | 00:00:00 | 13 rows

8. Danh sách khách hàng biết tên quốc gia: Mức 2:

use Northwind1

go

CREATE PROC DSKHBietQGMUC2(@QG nvarchar(15)) --Danh sách KH biết quốc gia mức 2

AS

BEGIN

IF (@QG=N'USA' OR @QG=N'UK')

SELECT * FROM Northwind1.dbo.KH1 WHERE Country=@QG

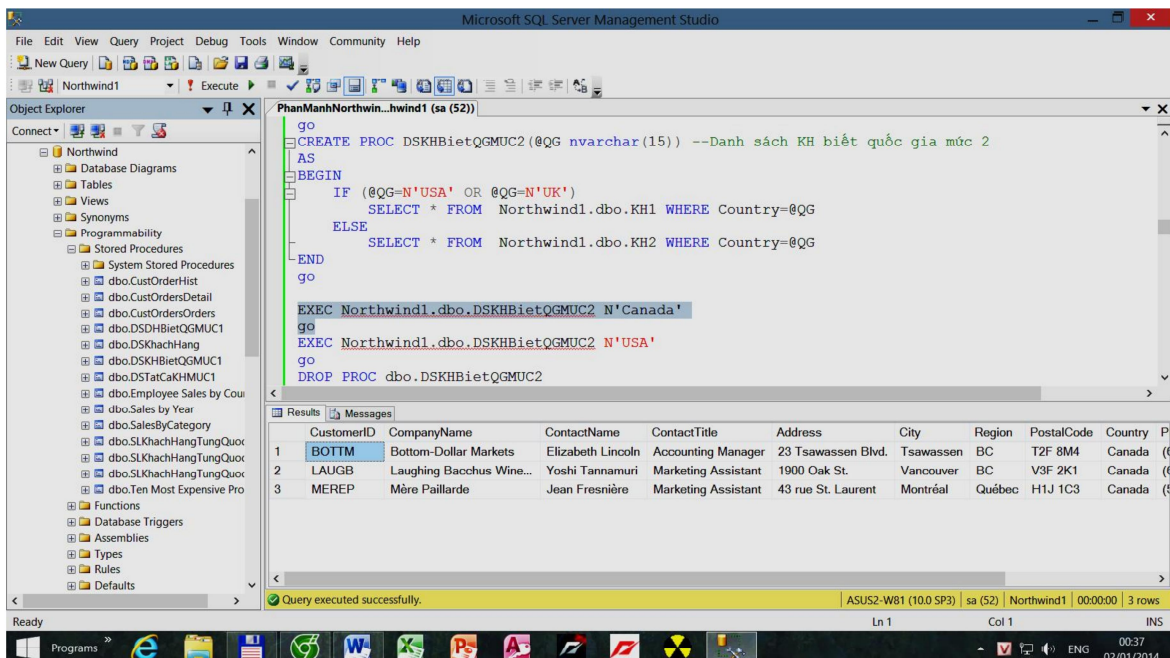
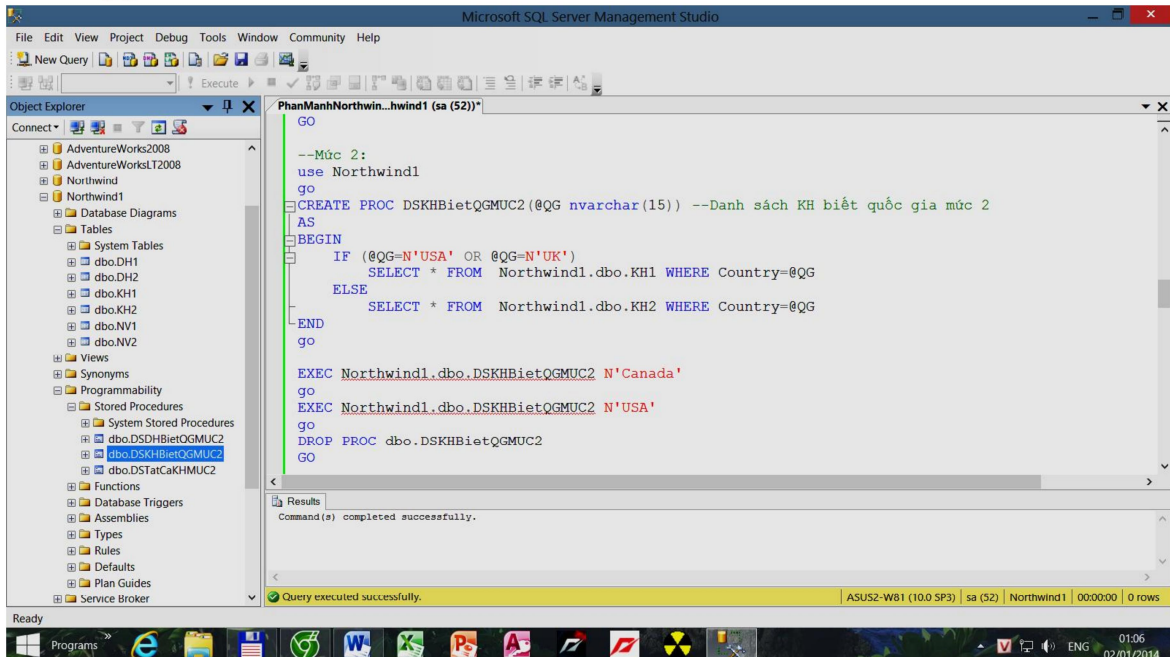
ELSE

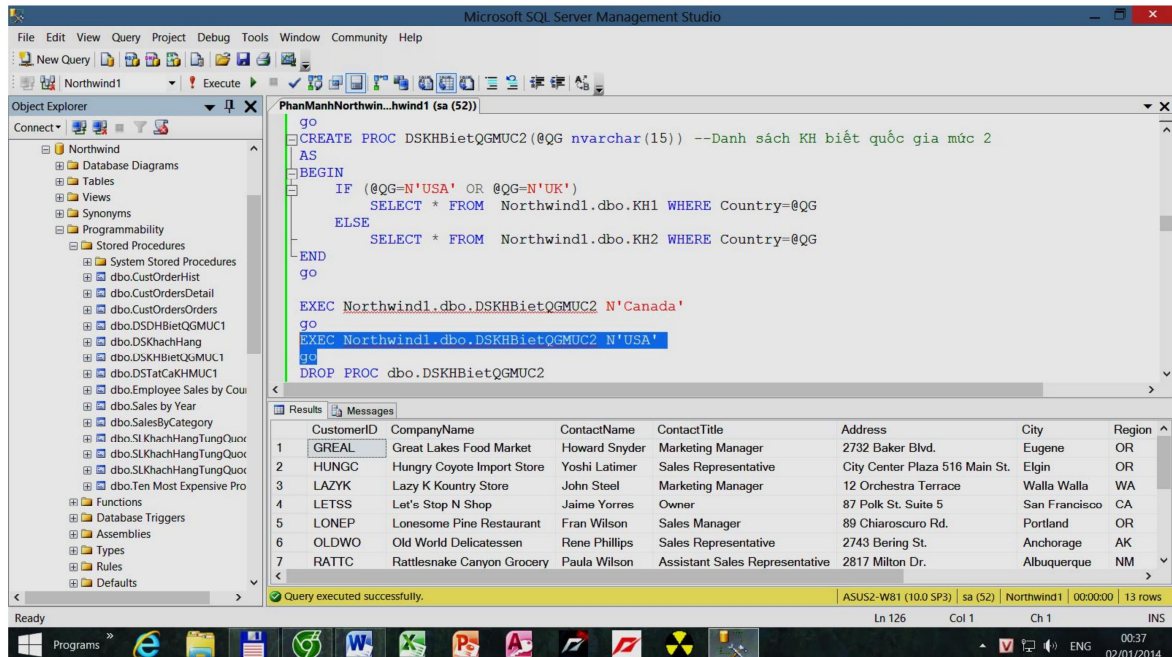
```

SELECT * FROM Northwind1.dbo.KH2 WHERE Country=@QG
END
go

EXEC Northwind1.dbo.DSKHBietQGMUC2 N'Canada'
go
EXEC Northwind1.dbo.DSKHBietQGMUC2 N'USA'
go
DROP PROC dbo.DSKHBietQGMUC2
GO

```





9. Phân mảnh ngang dẫn xuất: Phân mảnh bảng Đơn hàng dẫn xuất theo phân mảnh Khách hàng

-Tạo phân mảnh DH1 (đơn hàng 1) chứa các đơn hàng do các KH Mỹ và Anh mua

-Tạo phân mảnh DH2 (đơn hàng 2) chứa các đơn hàng do các KH còn lại mua

USE Northwind1

GO

--2 bước: Tạo thiết kế (bỏ IDENTITY cho cột khóa chính)
và lấy dữ liệu cho PM DH1 (đơn hàng 1)

--Tạo thiết kế DH1:

```

CREATE TABLE Northwind1.[dbo].[DH1] (
    [OrderID] [int] PRIMARY KEY,
    [CustomerID] [nchar] (5) NULL,
    [EmployeeID] [int] NULL,
    [OrderDate] [datetime] NULL,
    [RequiredDate] [datetime] NULL,
    [ShippedDate] [datetime] NULL,
    [ShipVia] [int] NULL,
    [Freight] [money] NULL,
    [ShipName] [nvarchar] (40) NULL,
    [ShipAddress] [nvarchar] (60) NULL,
    [ShipCity] [nvarchar] (15) NULL,
    [ShipRegion] [nvarchar] (15) NULL,
    [ShipPostalCode] [nvarchar] (10) NULL,
    [ShipCountry] [nvarchar] (15) NULL)
GO

```

--Lay DL cho PM DH1 từ PM KH1:

```

INSERT INTO dbo.DH1
    SELECT * FROM Northwind.dbo.Orders DH

```

```

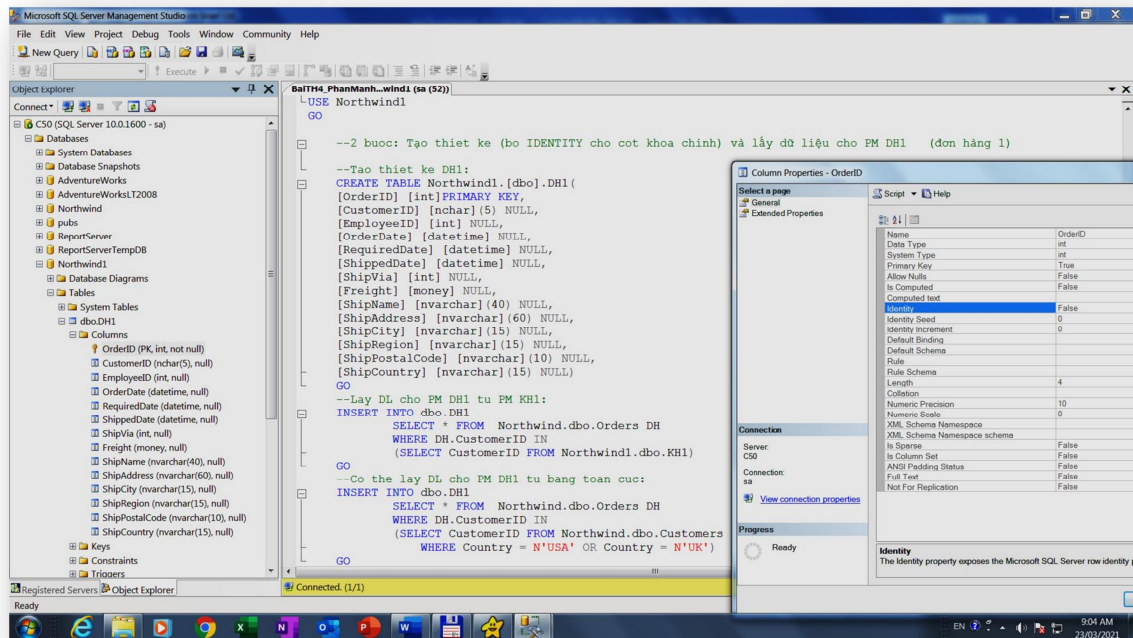
        WHERE DH.CustomerID IN
        (SELECT CustomerID FROM Northwind1.dbo.KH1)
GO
--Co the lay DL cho PM DH1 tu bang toan cuc:
INSERT INTO dbo.DH1
    SELECT * FROM Northwind.dbo.Orders DH
    WHERE DH.CustomerID IN
    (SELECT CustomerID FROM Northwind.dbo.Customers
        WHERE Country = N'USA' OR Country = N'UK')
GO
--Tao thiet ke DH2:
CREATE TABLE Northwind1.[dbo].DH2 (
    [OrderID] [int] PRIMARY KEY,
    [CustomerID] [nvarchar](5) NULL,
    [EmployeeID] [int] NULL,
    [OrderDate] [datetime] NULL,
    [RequiredDate] [datetime] NULL,
    [ShippedDate] [datetime] NULL,
    [ShipVia] [int] NULL,
    [Freight] [money] NULL,
    [ShipName] [nvarchar](40) NULL,
    [ShipAddress] [nvarchar](60) NULL,
    [ShipCity] [nvarchar](15) NULL,
    [ShipRegion] [nvarchar](15) NULL,
    [ShipPostalCode] [nvarchar](10) NULL,
    [ShipCountry] [nvarchar](15) NULL)
GO
--Lay DL cho PM DH2 tu PM KH2:
INSERT INTO dbo.DH2
    SELECT * FROM Northwind.dbo.Orders DH
    WHERE DH.CustomerID IN
    (SELECT CustomerID FROM Northwind1.dbo.KH2)
GO
--Co the lay DL cho PM DH2 tu bang toan cuc:
INSERT INTO dbo.DH2
    SELECT * FROM Northwind.dbo.Orders DH
    WHERE DH.CustomerID IN
    (SELECT CustomerID FROM Northwind.dbo.Customers
        WHERE Country <> N'USA' AND Country <>
N'UK')
GO

--Danh sách tất cả Đơn hàng mức 1
SELECT * FROM Northwind.dbo.Orders
GO

--Danh sách tất cả Đơn hàng mức 2
SELECT * FROM Northwind1.dbo.DH1
UNION
SELECT * FROM Northwind1.dbo.DH2

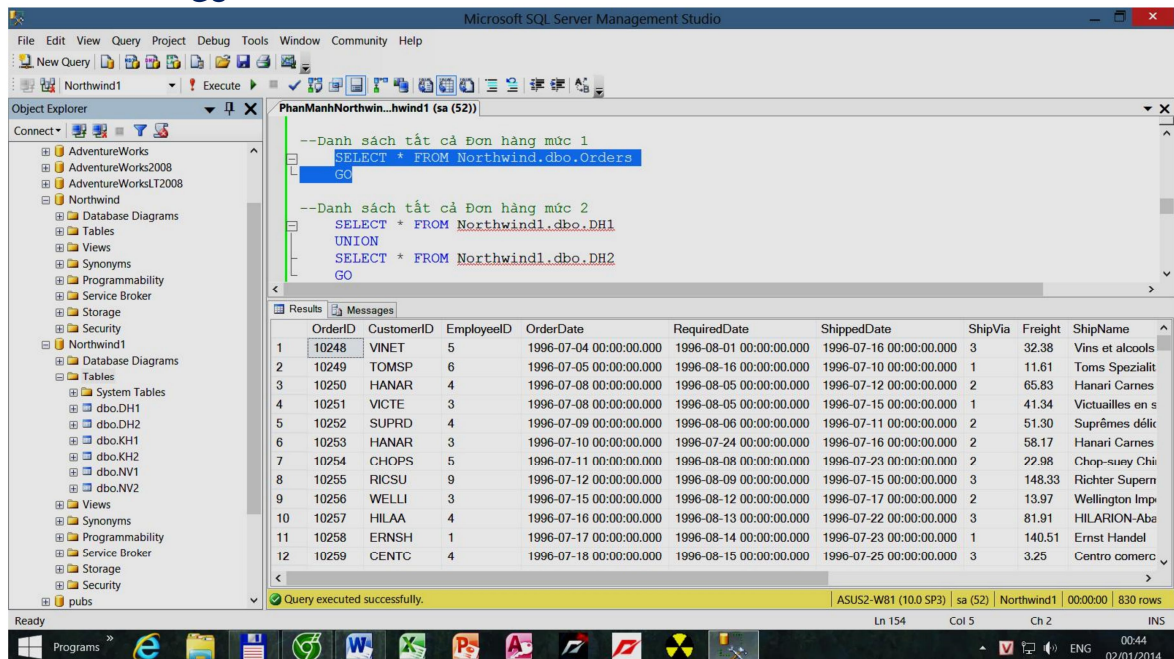
```

GO



10. Danh sách tất cả Đơn hàng mức 1

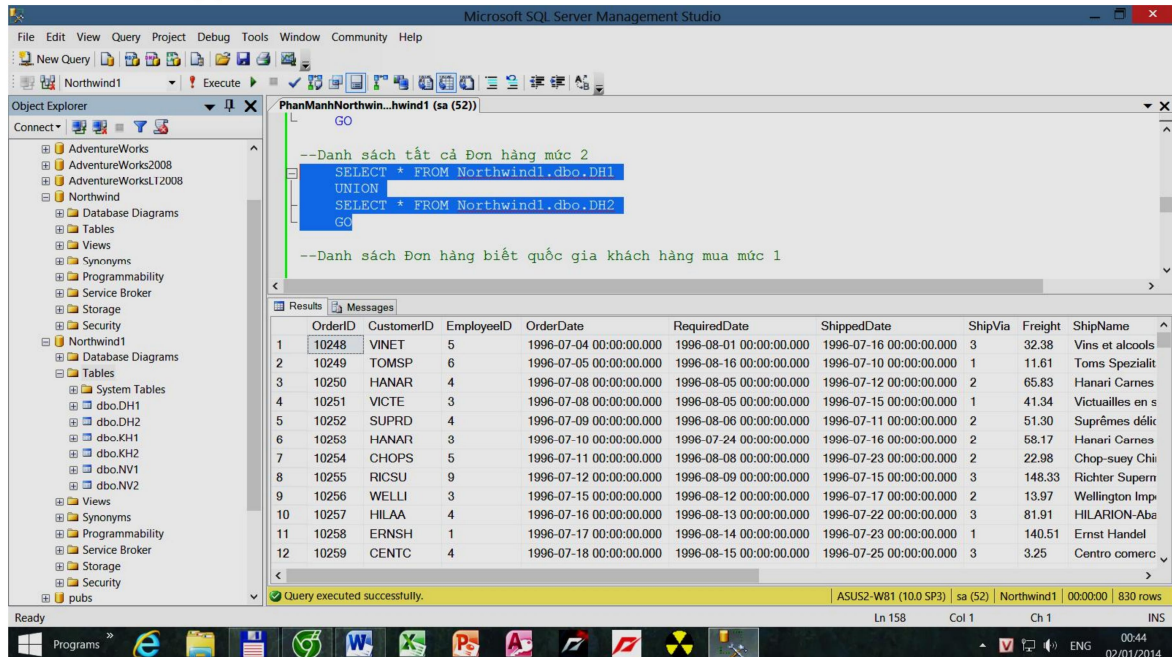
SELECT * FROM Northwind.dbo.Orders
GO



11. Danh sách tất cả Đơn hàng mức 2

SELECT * FROM Northwind1.dbo.DH1
UNION
SELECT * FROM Northwind1.dbo.DH2

GO



12. Danh sách Đơn hàng biết quốc gia khách hàng mua, mức 1:

```

use Northwind
go
CREATE PROC dbo.DSDHBietQGMUC1(@QG nvarchar(15))
AS
BEGIN
    SELECT * FROM Northwind.dbo.Orders DH
    WHERE DH.CustomerID IN
        (SELECT CustomerID FROM Northwind.dbo.Customers WHERE
        Country=@QG)
END
go

EXEC DSDHBietQGMUC1 N'Canada'
go
EXEC DSDHBietQGMUC1 N'USA'
go
DROP PROC DSDHBietQGMUC1
GO

```

Microsoft SQL Server Management Studio

Object Explorer: Northwind database structure.

Query Editor (PhanManhNorthwin...hwind (sa (52))):

```
--Danh sách Đơn hàng biết quốc gia khách hàng mua mức 1
use Northwind
go
CREATE PROC dbo.DSDHBietQGMUC1(@QG nvarchar(15)) --Danh sách DH biết quốc gia mua mức 1
AS
BEGIN
    SELECT * FROM Northwind.dbo.Orders DH
    WHERE DH.CustomerID IN
        (SELECT CustomerID FROM Northwind.dbo.Customers WHERE Country=@QG)
END
go
EXEC DSDHBietQGMUC1 N'Canada'
go
EXEC DSDHBietQGMUC1 N'USA'
go
DROP PROC DSDHBietQGMUC1
GO
```

Results:

	ShippedDate	ShipVia	Freight	ShipName	ShipAddress	ShipCity	ShipRegion	ShipPostalCode	ShipCountry
1	1996-10-21 00:00:00.000	2	52.84	Mère Paillarde	43 rue St. Laurent	Montréal	Québec	H1J 1C3	Canada
2	1996-11-04 00:00:00.000	2	15.66	Mère Paillarde	43 rue St. Laurent	Montréal	Québec	H1J 1C3	Canada
3	1996-12-13 00:00:00.000	2	20.39	Mère Paillarde	43 rue St. Laurent	Montréal	Québec	H1J 1C3	Canada
4	1996-12-24 00:00:00.000	2	47.42	Bottom-Dollar Markets	23 Tsawassen Blvd.	Tsawassen	BC	T2F 8M4	Canada
5	1997-01-15 00:00:00.000	3	2.40	Bottom-Dollar Markets	23 Tsawassen Blvd.	Tsawassen	BC	T2F 8M4	Canada
6	1997-01-21 00:00:00.000	3	23.65	Bottom-Dollar Markets	23 Tsawassen Blvd.	Tsawassen	BC	T2F 8M4	Canada

Query executed successfully. | ASUS2-W81 (10.0 SP3) | sa (52) | Northwind | 00:00:00 | 30 rows

Microsoft SQL Server Management Studio

Object Explorer: Northwind database structure.

Query Editor (PhanManhNorthwin...hwind (sa (52))):

```
--Danh sách Đơn hàng biết quốc gia khách hàng mua mức 1
use Northwind
go
CREATE PROC dbo.DSDHBietQGMUC1(@QG nvarchar(15)) --Danh sách DH biết quốc gia mua mức 1
AS
BEGIN
    SELECT * FROM Northwind.dbo.Orders DH
    WHERE DH.CustomerID IN
        (SELECT CustomerID FROM Northwind.dbo.Customers WHERE Country=@QG)
END
go
EXEC DSDHBietQGMUC1 N'Canada'
go
EXEC DSDHBietQGMUC1 N'USA'
go
DROP PROC DSDHBietQGMUC1
GO
```

Results:

	ShippedDate	ShipVia	Freight	ShipName	ShipAddress	ShipCity	ShipRegion	ShipPostalCode	ShipCountry
1	1996-10-21 00:00:00.000	2	52.84	Mère Paillarde	43 rue St. Laurent	Montréal	Québec	H1J 1C3	Canada
2	1996-11-04 00:00:00.000	2	15.66	Mère Paillarde	43 rue St. Laurent	Montréal	Québec	H1J 1C3	Canada
3	1996-12-13 00:00:00.000	2	20.39	Mère Paillarde	43 rue St. Laurent	Montréal	Québec	H1J 1C3	Canada
4	1996-12-24 00:00:00.000	2	47.42	Bottom-Dollar Markets	23 Tsawassen Blvd.	Tsawassen	BC	T2F 8M4	Canada
5	1997-01-15 00:00:00.000	3	2.40	Bottom-Dollar Markets	23 Tsawassen Blvd.	Tsawassen	BC	T2F 8M4	Canada
6	1997-01-21 00:00:00.000	3	23.65	Bottom-Dollar Markets	23 Tsawassen Blvd.	Tsawassen	BC	T2F 8M4	Canada

Query executed successfully. | ASUS2-W81 (10.0 SP3) | sa (52) | Northwind | 00:00:00 | 122 rows

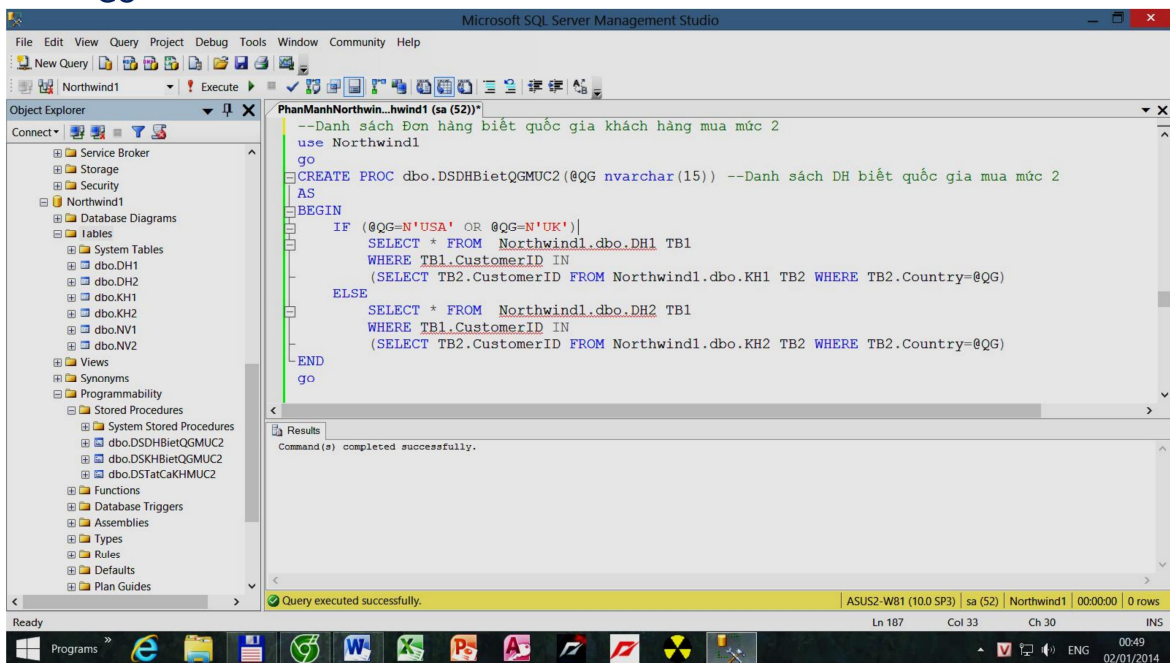
13. --Danh sách Đơn hàng biết quốc gia khách hàng mua mức 2

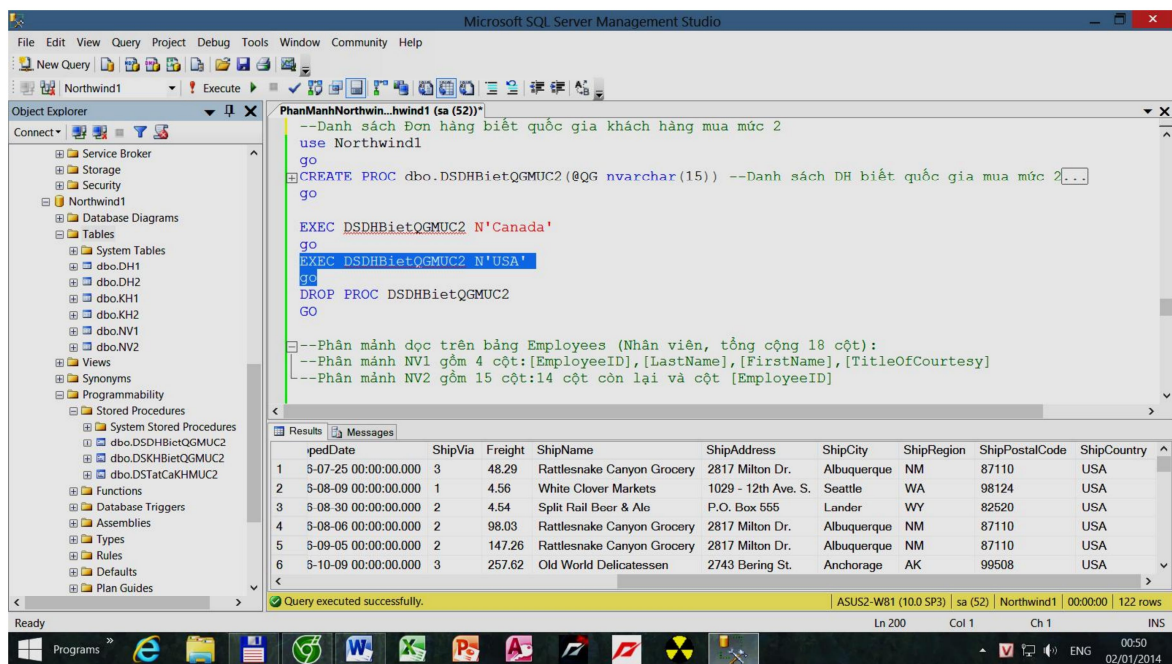
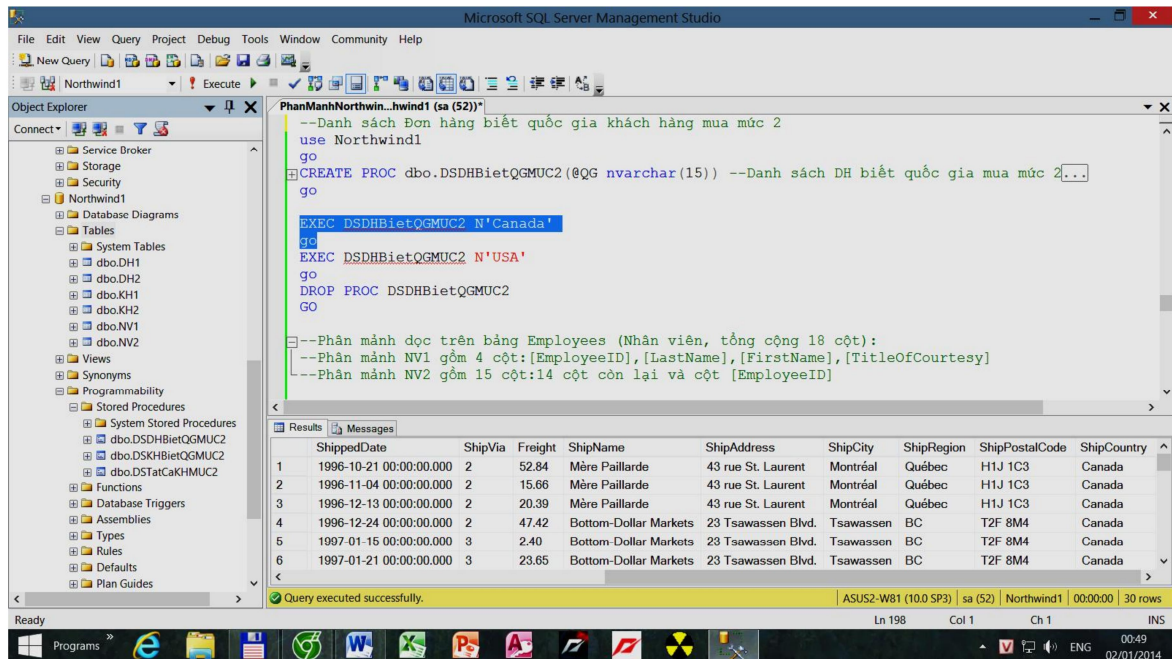
```
use Northwind1
go
CREATE PROC dbo.DSDHBietQGMUC2(@QG nvarchar(15)) --Danh sách DH
biết quốc gia mua mức 2
AS
BEGIN
    IF (@QG=N'USA' OR @QG=N'UK')
        SELECT * FROM Northwind1.dbo.DH1 TB1
```

```

WHERE TB1.CustomerID IN
      (SELECT TB2.CustomerID FROM Northwind1.dbo.KH1 TB2
WHERE TB2.Country=@QG)
ELSE
      SELECT * FROM Northwind1.dbo.DH2 TB1
      WHERE TB1.CustomerID IN
      (SELECT TB2.CustomerID FROM Northwind1.dbo.KH2 TB2
WHERE TB2.Country=@QG)
END
go
EXEC DSDHBietQGMUC2 N'Canada'
go
EXEC DSDHBietQGMUC2 N'USA'
go
DROP PROC DSDHBietQGMUC2
GO

```





Phân mảnh dọc bảng dữ liệu:

14. Phân mảnh dọc trên bảng Employees (Nhân viên, tổng cộng 18 cột):

-Phân mảnh NV1 gồm 4 cột:[EmployeeID], [LastName], [FirstName], [TitleOfCourtesy]

-Phân mảnh NV2 gồm 15 cột:14 cột còn lại và cột [EmployeeID]

USE Northwind1

GO

--2 bước: Tạo thiết kế PM NV1 và lấy dữ liệu cho phân mảnh NV1(NHÂN VIÊN 1)

--Tạo PM NV1:

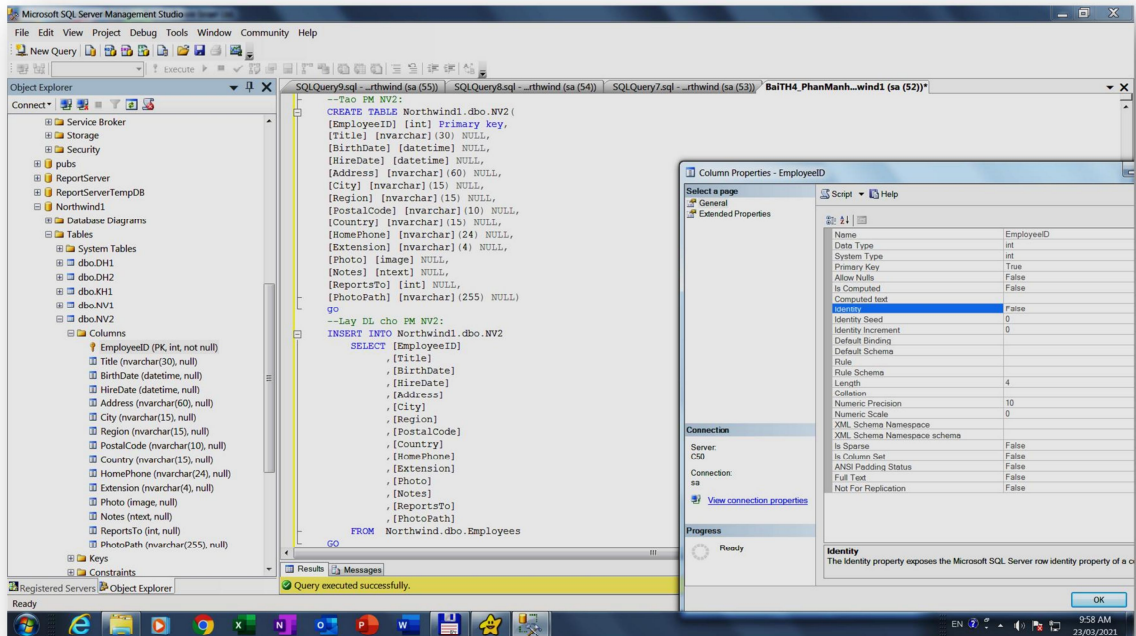
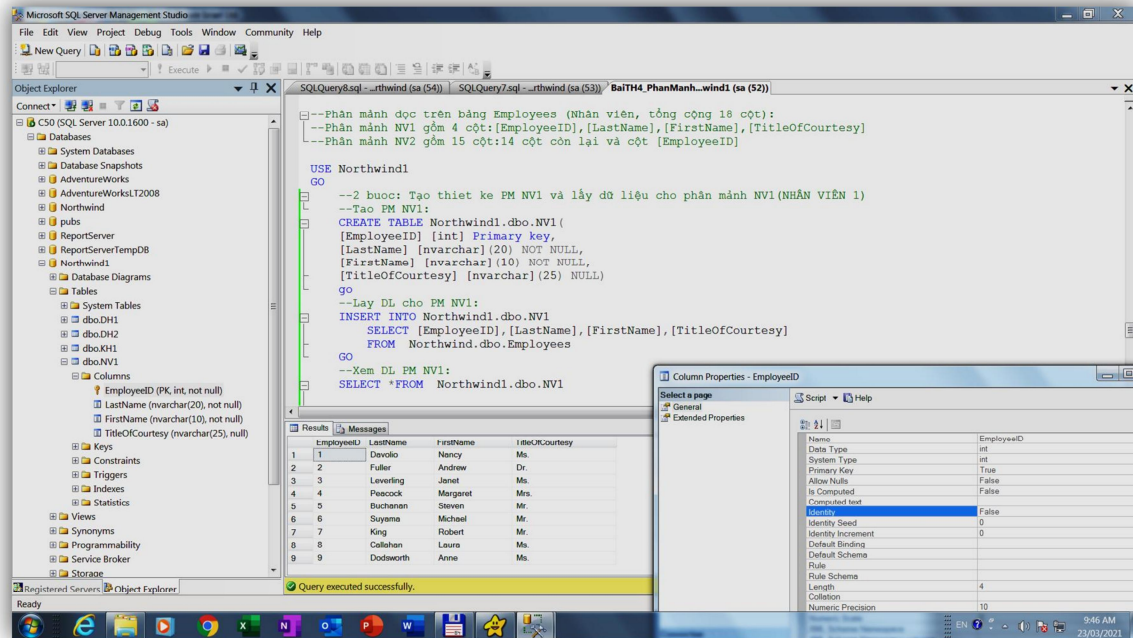
CREATE TABLE Northwind1.dbo.NV1 (

```

[EmployeeID] [int] Primary key,
[LastName] [nvarchar](20) NOT NULL,
[FirstName] [nvarchar](10) NOT NULL,
[TitleOfCourtesy] [nvarchar](25) NULL)
go
--Lay DL cho PM NV1:
INSERT INTO Northwind1.dbo.NV1
    SELECT [EmployeeID],[LastName],[FirstName],[TitleOfCourtesy]
    FROM Northwind.dbo.Employees
GO
--Xem DL PM NV1:
SELECT *FROM Northwind1.dbo.NV1

-----
--Tao PM NV2:
CREATE TABLE Northwind1.dbo.NV2 (
[EmployeeID] [int] Primary key,
[Title] [nvarchar](30) NULL,
[BirthDate] [datetime] NULL,
[HireDate] [datetime] NULL,
[Address] [nvarchar](60) NULL,
[City] [nvarchar](15) NULL,
[Region] [nvarchar](15) NULL,
[PostalCode] [nvarchar](10) NULL,
[Country] [nvarchar](15) NULL,
[HomePhone] [nvarchar](24) NULL,
[Extension] [nvarchar](4) NULL,
[Photo] [image] NULL,
[Notes] [ntext] NULL,
[ReportsTo] [int] NULL,
[PhotoPath] [nvarchar](255) NULL)
go
--Lay DL cho PM NV2:
INSERT INTO Northwind1.dbo.NV2
    SELECT [EmployeeID]
        , [Title]
        , [BirthDate]
        , [HireDate]
        , [Address]
        , [City]
        , [Region]
        , [PostalCode]
        , [Country]
        , [HomePhone]
        , [Extension]
        , [Photo]
        , [Notes]
        , [ReportsTo]
        , [PhotoPath]
    FROM Northwind.dbo.Employees
GO
--Xem DL PM NV2:
SELECT *FROM Northwind1.dbo.NV2

```



15. Danh sách tất cả nhân viên, mức 1 (9 hàng, 18 cột):
SELECT * FROM Northwind.dbo.Employees
go

PhanManhNorthwin...hwind (sa (52))

```
-- Danh sách tất cả nhân viên mức 1 (9 hàng, 18 cột):
SELECT * FROM Northwind.dbo.Employees
go

-- Danh sách tất cả nhân viên mức 2 (9 hàng, 18 cột):
SELECT TB1.[EmployeeID]
, [LastName]
, [FirstName]
, [Title]
, [TitleOfCourtesy]
, [BirthDate]
, [HireDate]
, [Address]
```

EmployeeID	LastName	FirstName	Title	TitleOfCourtesy	BirthDate	HireDate	Address
1	Davolio	Nancy	Sales Representative	Ms.	1948-12-08 00:00:00.000	1992-05-01 00:00:00.000	507 - 20th Ave. E. Ap
2	Fuller	Andrew	Vice President, Sales	Dr.	1952-02-19 00:00:00.000	1992-08-14 00:00:00.000	908 W. Capital Way
3	Leverling	Janet	Sales Representative	Ms.	1963-08-30 00:00:00.000	1992-04-01 00:00:00.000	722 Moss Bay Blvd.
4	Peacock	Margaret	Sales Representative	Mrs.	1937-09-19 00:00:00.000	1993-05-03 00:00:00.000	4110 Old Redmond Rd
5	Buchanan	Stevon	Sales Manager	Mr.	1955-03-04 00:00:00.000	1993-10-17 00:00:00.000	14 Garrett Hill
6	Suyama	Michael	Sales Representative	Mr.	1963-07-02 00:00:00.000	1993-10-17 00:00:00.000	Coventry House Mine
7	King	Robert	Sales Representative	Mr.	1960-05-29 00:00:00.000	1994-01-02 00:00:00.000	Edgeham Hollow Win
8	Callahan	Laura	Inside Sales Coordi...	Ms.	1958-01-09 00:00:00.000	1994-03-05 00:00:00.000	4726 - 11th Ave. N.E.
9	Dodsw...	Anne	Sales Representative	Ms.	1966-01-27 00:00:00.000	1994-11-15 00:00:00.000	7 Houndstooth Rd.

Ready | Query executed successfully. | ASUS2-W81 (10.0 SP3) | sa (52) | Northwind | 00:00:00 | 9 rows

16. Danh sách tất cả nhân viên mức 2 (9 hàng, 18 cột):

```
SELECT TB1.[EmployeeID]
, [LastName]
, [FirstName]
, [Title]
, [TitleOfCourtesy]
, [BirthDate]
, [HireDate]
, [Address]
, [City]
, [Region]
, [PostalCode]
, [Country]
, [HomePhone]
, [Extension]
, [Photo]
, [Notes]
, [ReportsTo]
, [PhotoPath]
FROM Northwind1.dbo.NV1 TB1, Northwind1.dbo.NV2 TB2
WHERE TB1.EmployeeID=TB2.EmployeeID
go
```

Microsoft SQL Server Management Studio

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Northwind1

Object Explorer

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```
-- Danh sách tất cả nhân viên mức 2 (9 hàng, 18 cột):
SELECT TB1.[EmployeeID]
      ,[LastName],[FirstName],[Title]
      ,[TitleOfCourtesy],[BirthDate]
      ,[HireDate],[Address],[City],[Region]
      ,[PostalCode],[Country],[HomePhone]
      ,[Extension],[Photo],[Notes],[ReportsTo],[PhotoPath]
FROM Northwind1.dbo.NV1 TB1,Northwind1.dbo.NV2 TB2
WHERE TB1.EmployeeID=TB2.EmployeeID
go
```

Results Messages

EmployeeID	LastName	FirstName	Title	TitleOfCourtesy	BirthDate	HireDate	Address
1	Davolio	Nancy	Sales Representative	Ms.	1948-12-08 00:00:00.000	1992-05-01 00:00:00.000	507 - 20th Ave. E.
2	Fuller	Andrew	Vice President, Sales	Dr.	1952-02-19 00:00:00.000	1992-08-14 00:00:00.000	908 W. Capital W
3	Leverling	Janet	Sales Representative	Ms.	1963-08-30 00:00:00.000	1992-04-01 00:00:00.000	722 Moss Bay Bl
4	Peacock	Margaret	Sales Representative	Mrs.	1937-09-19 00:00:00.000	1993-05-03 00:00:00.000	4110 Old Redmon
5	Buchanan	Steven	Sales Manager	Mr.	1955-03-04 00:00:00.000	1993-10-17 00:00:00.000	14 Garrett Hill
6	Suyama	Michael	Sales Representative	Mr.	1963-07-02 00:00:00.000	1993-10-17 00:00:00.000	Coventry House I
7	King	Robert	Sales Representative	Mr.	1960-05-29 00:00:00.000	1994-01-02 00:00:00.000	Edgeham Hollow
8	Callahan	Laura	Inside Sales Coordinator	Ms.	1958-01-09 00:00:00.000	1994-03-05 00:00:00.000	4726 - 11th Ave. N
9	Dodsworth	Anne	Sales Representative	Ms.	1966-01-27 00:00:00.000	1994-11-15 00:00:00.000	7 Houndstooth Rd

Query executed successfully.

ASUS2-WB1 (10.0 SP3) | sa (52) | Northwind1 | 00:00:00 | 9 rows

Ln 241 Col 59 Ch 59 00:58 02/01/2014

(HẾT PHẦN 1)